

Curriculum Vitae

Rishish Mishra

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Summary

Ph.D. in Mechanical Engineering from the Indian Institute of Technology Jammu, with a thesis on the hydrodynamics of microswimmers near fluid-fluid interfaces. Holds a Master's degree in Mechatronics Engineering and a Bachelor's degree in Mechanical Engineering. Has four years of research experience studying microswimmers and their interaction in multi-phase system. Currently focused on computational studies of multiphase flows involving droplets, active particles, and colloidal particles near fluid interfaces.

Education

PhD: Mechanical Engineering, IIT Jammu, India, 8.75 CGPA	2021–2026
M.Tech: Mechatronics, Centre for Advanced Studies, Lucknow, India, 9.18 CGPA	2018–2020
B.Tech: Mechanical Engineering, Institute of Engineering and Technology, Ayodhya, India, 72.54%	2012–2016

Areas of Interest

- Fluid Mechanics, Multiphase flows, CFD, LBM, Microswimmers, Low-Reynolds-number flow, and Fluid-Fluid interface.

Expertise

- *Mathematical Modeling*
- *Programming skills: MATLAB and Python*
- *Softwares: Ludwig (An open source lattice Boltzmann code), Paraview, Shell scripting, SolidWorks, and ANSYS Fluent.*
- *High performance computing (HPC)*

Publications

1. Mishra, R., Pothukuchi, H., Gidituri, H., & Lintuvuori, J. (2026). *Interface crossing behavior of prolate microswimmers: Thermo and hydrodynamics*. Physical Review Fluids, **11**(1), 014002.
2. Mishra, R., Gosika, M., Gidituri, H., & Pothukuchi, H. (2024). *Flow structure around a microswimmer at a fluid-fluid interface*. Fluid Dynamics Research, **56**(4), 045507.
3. Mishra, R., Dubey, V., Singh, R. K., Sharma, A. K., & Katiyar, J. K. (2022). *A novel conjugate heat transfer approach to determine the temperature distribution in a single-point cutting tool under different conditions*. Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering, 09544089221125633.
4. Shukla, K., Mishra, R., & Sarkar, P. (2021). *Understanding soilless engineered soil as a sustainable growing material for food production in a green roof*. Materials Today: Proceedings, **43**, 3054–3060.

Manuscript under preparation and Review

1. Mishra, R., Pothukuchi, H., Gidituri, H., & Lintuvuori, J. *To be Trapped or not, dynamics of spherical microswimmer near a fluid-fluid interface.*
2. Mishra, R., Pothukuchi, H., & Gidituri, H. *Pair-wise interaction between a pair of pusher-type bacterium at the fluid-fluid interface.*

Workshops, and Schools Attended

1. Interfacial Transport Phenomenon and Solution Technique, GIAN. January 13-17, 2023.
2. Simulating energy materials with ESPResSO and waLBerla, CECAM Workshop at Institute of Physics, University of Stuttgart, Germany.
3. Challenges in the Physics of Active and Biological Matter, Aalto University, Finland.

Conferences and Poster Presented

1. “Dynamics of a prolate microswimmer near a fluid-fluid interface”. CompFlu 2025, Complex Fluids Symposium 2025. Indian Institute of Science, Bangalore.
2. “Pairwise Interaction between a pair of pushers at the interface”. International soft matter conference 2025, Crete Greece.
3. “Pairwise Interaction of microswimmers at the interface”. CompFlu 2024, Complex Fluids Symposium 2024. Indian Institute of Madras.
4. “Interfacial Pusher Bacterium”. CompFlu 2023, Complex Fluids Symposium 2023. Indian Institute of Technology Hyderabad.

Awards and Grants

1. International Travel Support, Anusandhan National Research Foundation (ANRF), India, 2025.
2. Qualified Graduate Aptitude Test in Engineering (GATE), Ministry of Education, India, February 2018.

Work Experience

1. Junior Research Fellow at IIT Ropar (Oct 2020 -Jan 2021) Mechanical Engineering department (Micro-green roofing, DST-funded SERB Project)
2. A 4-week industrial internship at Entuple Technologies Pvt. Ltd., Bangalore, with hands-on training in ANSYS Workbench, including CFD, Structural, and FSI modules

Teaching Experience

1. Teaching Assistant, “Numerical Methods”, IIT Jammu, July-November 2024.
2. Teaching Assistant, “Software Tools”, IIT Jammu, January-May 2024.
3. Teaching Assistant, “Thermal and Fluid Laboratory”, IIT Jammu, July-November 2023.
4. Teaching Assistant, “Numerical Methods”, IIT Jammu, January-May 2023.
5. Teaching Assistant, “Engineering Thermodynamics”, IIT Jammu, July-November 2022.

Recommendations

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